

Introduction

"The objective of this project is to implement network-level content filtering using pfSense to restrict access to unauthorized websites."

System Environment

"The network setup consists of a pfSense firewall managing traffic and a SliTaz Linux client machine."

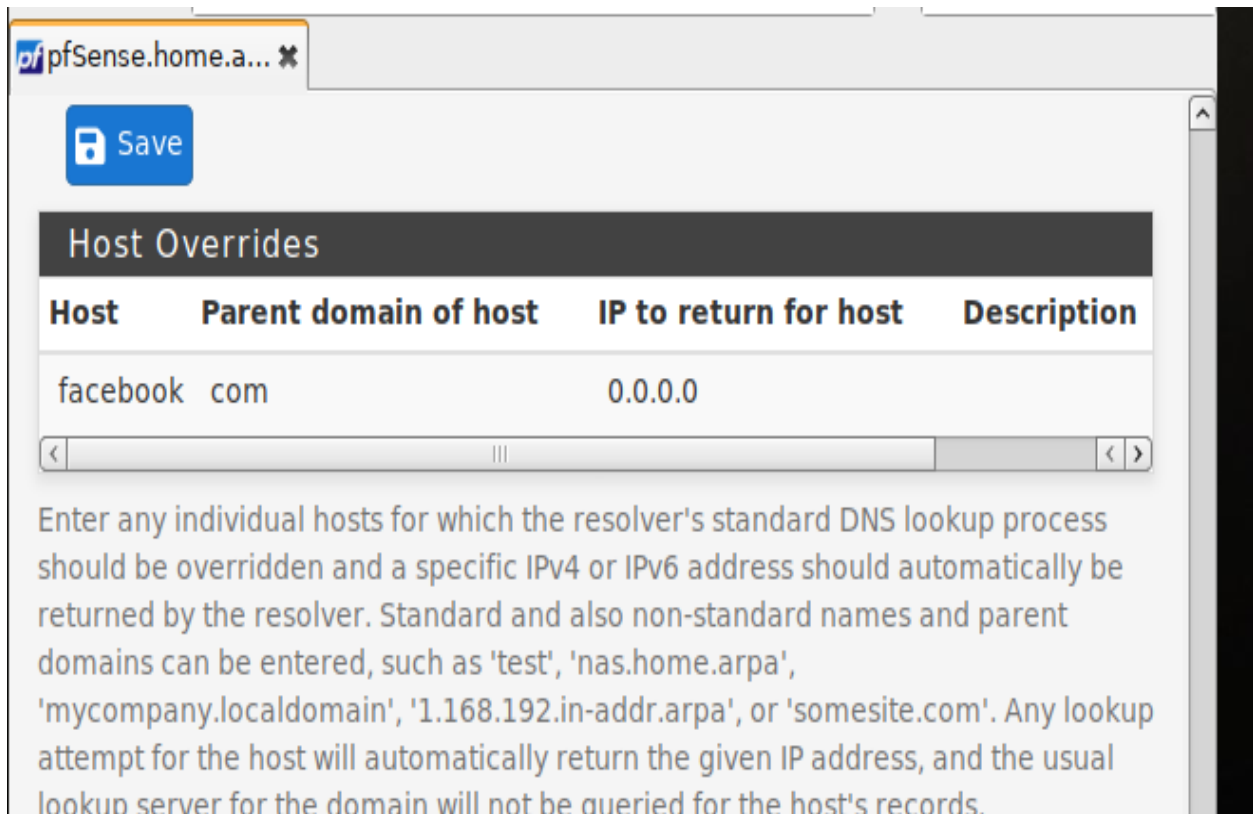
Implementation Steps

Caption:

Figure 1: DNS Resolver Host Overrides configuration in pfSense.

Description:

"This configuration illustrates the implementation of network-level domain filtering. By adding a Host Override for 'facebook.com' and setting the IP address to '0.0.0.0', all DNS queries for this domain are effectively nullified. This prevents the domain from resolving to a valid IP address, thereby blocking access to the website."



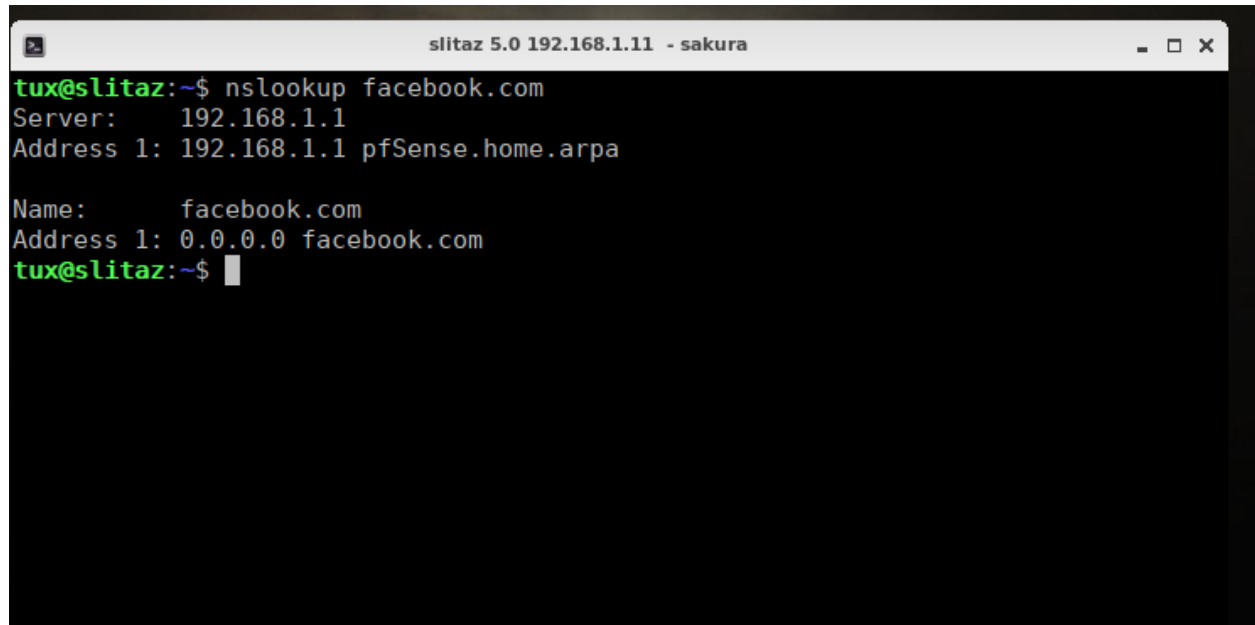
Verification and Testing

Caption:

Figure 2: Verifying the DNS block using the nslookup command.

Description:

"To verify the configuration, the 'nslookup' command was executed from the client machine. The output, showing 'Address 1: 0.0.0.0', confirms that the pfSense DNS Resolver successfully intercepted the request and returned a non-routable address instead of the actual web server IP, indicating a successful implementation of the block."

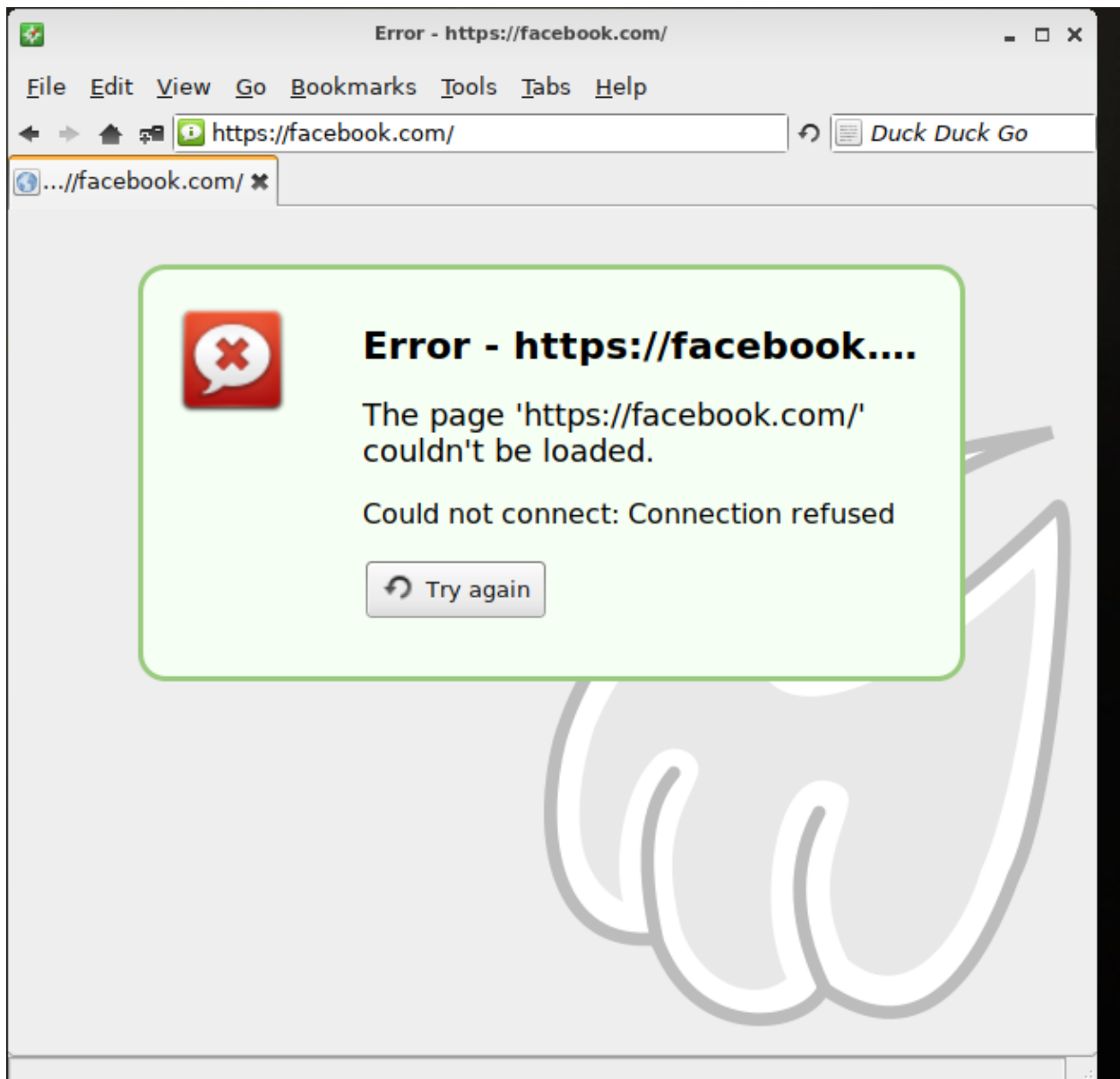
A terminal window titled "slitaz 5.0 192.168.1.11 - sakura" with standard window controls. The prompt is "tux@slitaz:~\$". The command "nslookup facebook.com" has been executed. The output shows the DNS server as 192.168.1.1 and the address for pfSense.home.arpa as 192.168.1.1. For facebook.com, the name is resolved to 0.0.0.0. The prompt returns to "tux@slitaz:~\$".

```
tux@slitaz:~$ nslookup facebook.com
Server:      192.168.1.1
Address 1:   192.168.1.1 pfSense.home.arpa

Name:        facebook.com
Address 1:   0.0.0.0 facebook.com
tux@slitaz:~$
```

Caption: Figure 3: Browser response showing blocked access to the domain.

Description: "When attempting to access the domain via HTTPS, the browser fails to establish a secure connection because the DNS resolution points to a non-routable IP address (0.0.0.0). This verifies that the content filtering policy effectively blocks the website at the browser level."



Project Conclusion :

"The implementation of DNS-based content filtering using pfSense provides a robust mechanism to manage and restrict internet access across the local network. By overriding specific domain lookups at the DNS level, unauthorized access to restricted websites is

prevented before the connection is established. The testing results confirm that this configuration is effective, scalable, and easy to manage.